

Experiment 3: Google Cloud CLI and App Engine Project Setup

Aim

To install the modern Google Cloud CLI, initialize a Google Cloud project, install the App Engine Python component, and prepare a simple Python web application.

Tools Required

- Ubuntu VM
- Python 3
- Google Cloud CLI
- Internet connection
- Google account / Google Cloud project

Detailed Procedure

Open Terminal and verify Python 3 is available.

```
python3 --version
```

Install Google Cloud CLI using the current Linux installation package/method. If using the archive method from the lab, download, extract, and run the installer.

```
curl -O https://dl.google.com/dl/cloudsdk/channels/rapid/downloads/google-cloud-cli-linux-x86_64.tar.gz
tar -xf google-cloud-cli-linux-x86_64.tar.gz
./google-cloud-sdk/install.sh
```

Initialize the CLI and sign in.

```
gcloud init
```

During initialization, sign in with the Google account and select the required Google Cloud project.

Install the App Engine Python component.

```
gcloud components install app-engine-python
```

Create the application directory.

```
mkdir gae-helloworld
cd gae-helloworld
```

Create main.py with the following code.

```
from flask import Flask

app = Flask(__name__)

@app.route("/")
def hello():
    return "Hello world from Google App Engine!"

if __name__ == "__main__":
    app.run(host="127.0.0.1", port=8080, debug=True)
```

Create app.yaml. Use a currently supported Python runtime; in the completed workflow Python 3.14 was used.

```
runtime: python314  
handlers:  
- url: /*  
  script: auto
```

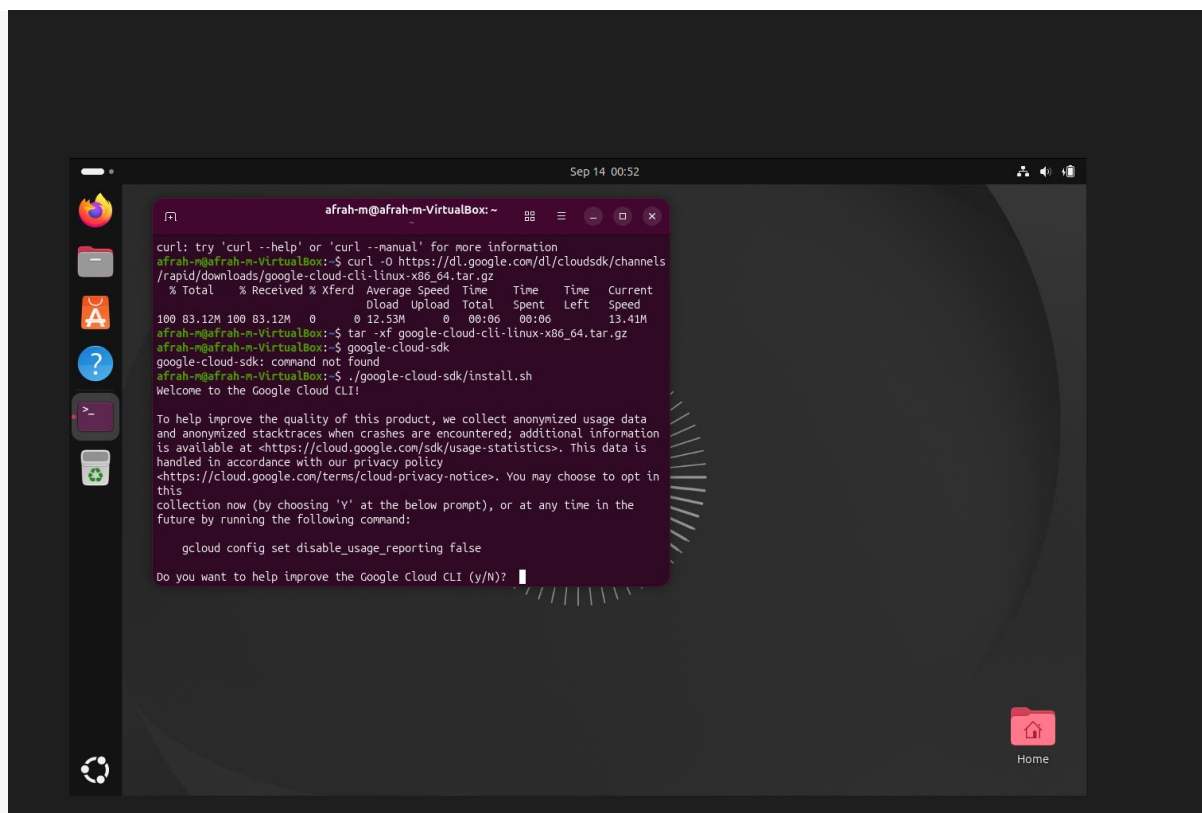
Create requirements.txt. The completed workflow used Flask 3.1.3 for compatibility with the current Python environment.

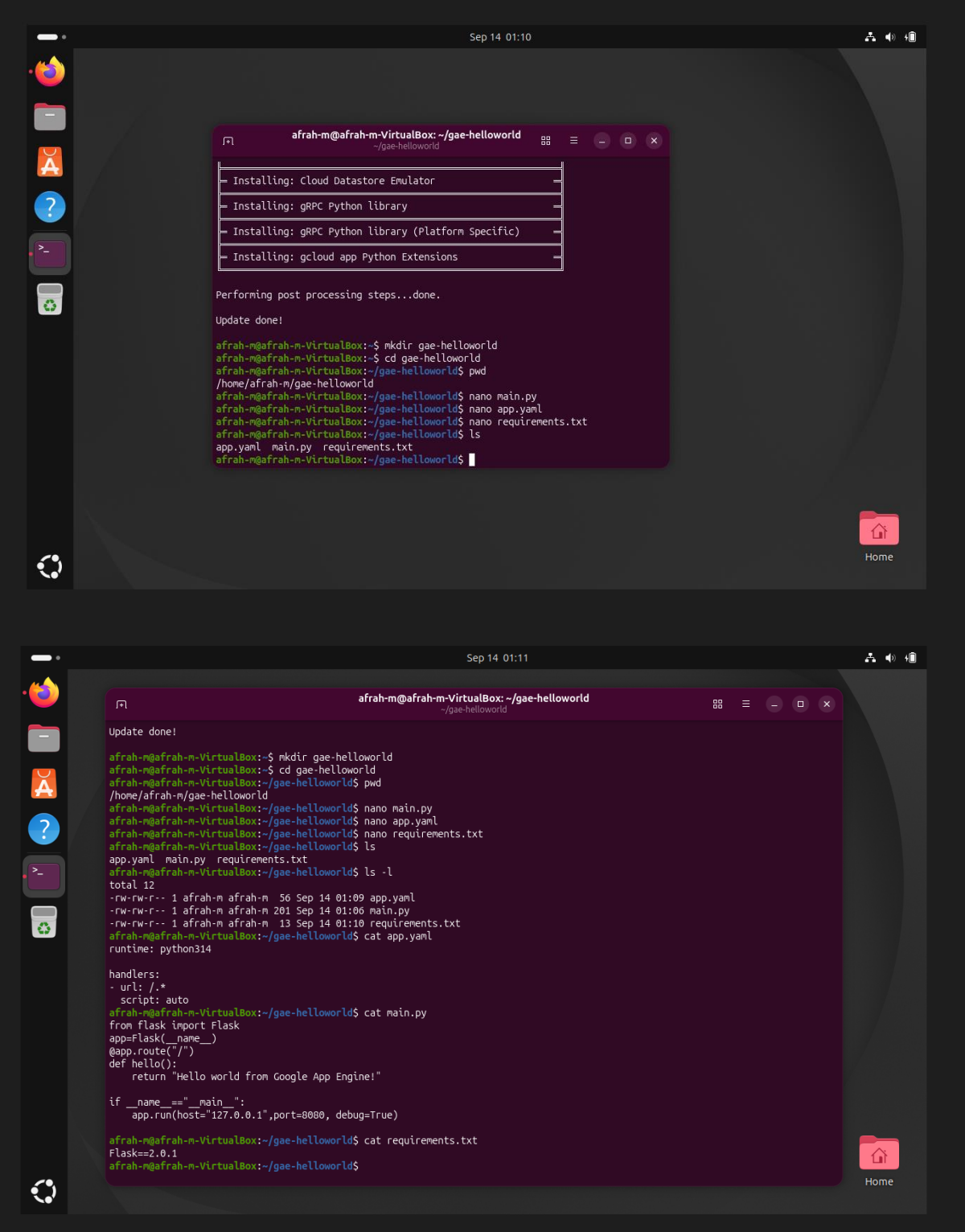
```
Flask==3.1.3
```

Important Compatibility Note

The original lab sheet specifies python39 and Flask 2.0.1. Those versions caused compatibility problems in the current Ubuntu/Python environment. The completed procedure therefore uses Python 3.14 in app.yaml and Flask 3.1.3.

Output





The image displays two sequential screenshots of a terminal window within a virtual machine environment, showing the steps to initialize a Google App Engine Python project.

Top Screenshot: The terminal window title is "afrah-m@afrah-m-VirtualBox: ~/gae-helloworld". It shows the installation of the Cloud Datastore Emulator, gRPC Python library, and gcloud app Python Extensions. After performing post-processing steps, the user creates a directory "gae-helloworld", navigates to it, and creates three files: "main.py", "app.yaml", and "requirements.txt".

```
afrah-m@afrah-m-VirtualBox: ~/gae-helloworld
- Installing: Cloud Datastore Emulator
- Installing: gRPC Python library
- Installing: gRPC Python library (Platform Specific)
- Installing: gcloud app Python Extensions

Performing post processing steps...done.
Update done!

afrah-m@afrah-m-VirtualBox:~$ mkdir gae-helloworld
afrah-m@afrah-m-VirtualBox:~$ cd gae-helloworld
afrah-m@afrah-m-VirtualBox:~/gae-helloworld$ pwd
/home/afrah-m/gae-helloworld
afrah-m@afrah-m-VirtualBox:~/gae-helloworld$ nano main.py
afrah-m@afrah-m-VirtualBox:~/gae-helloworld$ nano app.yaml
afrah-m@afrah-m-VirtualBox:~/gae-helloworld$ nano requirements.txt
afrah-m@afrah-m-VirtualBox:~/gae-helloworld$ ls
app.yaml  main.py  requirements.txt
afrah-m@afrah-m-VirtualBox:~/gae-helloworld$
```

Bottom Screenshot: The terminal window title is "afrah-m@afrah-m-VirtualBox: ~/gae-helloworld". It shows the user listing files, checking file permissions, and displaying the contents of "app.yaml" and "main.py".

```
afrah-m@afrah-m-VirtualBox:~/gae-helloworld$ ls
app.yaml  main.py  requirements.txt
afrah-m@afrah-m-VirtualBox:~/gae-helloworld$ ls -l
total 12
-rw-rw-r-- 1 afrah-m afrah-m 56 Sep 14 01:09 app.yaml
-rw-rw-r-- 1 afrah-m afrah-m 201 Sep 14 01:06 main.py
-rw-rw-r-- 1 afrah-m afrah-m 13 Sep 14 01:10 requirements.txt
afrah-m@afrah-m-VirtualBox:~/gae-helloworld$ cat app.yaml
runtime: python314

handlers:
- url: /*
  script: auto
afrah-m@afrah-m-VirtualBox:~/gae-helloworld$ cat main.py
from flask import Flask
app=Flask(__name__)
@app.route("/")
def hello():
    return "Hello world from Google App Engine!"

if __name__=="__main__":
    app.run(host="127.0.0.1",port=8080, debug=True)
afrah-m@afrah-m-VirtualBox:~/gae-helloworld$ cat requirements.txt
Flask==2.0.1
afrah-m@afrah-m-VirtualBox:~/gae-helloworld$
```

Result

Google Cloud CLI was initialized successfully and the App Engine Python project structure was prepared using a currently compatible Python runtime and Flask version.